

SKANDA VASISHTA

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EDUCATION

University of California, Santa Barbara

BS Computer Science, GPA: 3.87

MS Computer Science

Santa Barbara, CA

Sep 2022 - December 2025

January 2026 - December 2026

Relevant Courses: Data Structures/Algorithms, Operating Systems, Databases, Artificial Intelligence, Deep Learning, Networks, Compilers, Computer Architecture, Object-Oriented Programming

Awards/Honors: 8x Engineering Dean's Honors List, Engineering Honors

EXPERIENCE

Software Engineering Intern

Pacific Life

Newport Beach, CA

June 2025 - Present

- Architected economic scenario generation tool to streamline quantitative analysis workflows utilizing React/TypeScript frontend and AWS serverless backend, reducing manual processing time by 60%.
- Containerizing simulation engine with Docker, codifying infrastructure with Terraform, and implementing SQS-based asynchronous processing to enable parallel workload execution.
- Delivering features in Agile sprints via Azure DevOps, collaborating closely with quantitative analysts and fellow engineers on planning, code reviews, and CI/CD.

Machine Learning Researcher

UCSB NLP Group

Isla Vista, CA

April 2025 - Present

- Researching applications of multi-modal assistants for physical tasks under Prof. Xifeng Yan.
- Utilizing OpenCV, pre-trained HuggingFace Transformer models, and OpenAI API to create video processing pipeline and generate real-time feedback.
- Fine-tuning vision models to recognize task correctness and deliver instant corrective guidance.

Software Development Intern

NOAA Channel Islands

Santa Barbara, CA

June 2024 - February 2025

- Developed end-to-end full-stack project utilizing PHP for backend systems and API development, JavaScript for interactive frontend components, and MySQL for database systems.
- Integrated REST APIs to develop custom plugins, reducing API response times by 10% through caching.
- Integrated retrieval augmented generation chatbot into application utilizing Python, Langchain, and Chroma vector embeddings.

Undergraduate Researcher

UCSB Archlab

Isla Vista, CA

January 2024 - April 2025

- Prototyped and benchmarked subgraph isomorphism algorithms and heuristics using Python circuit-modeling libraries along with Pandas, Matplotlib, and Numpy.
- Researching applications of graph neural networks utilizing PyTorch PyG library for computer architecture applications.

LEADERSHIP

Gauche Sports Analytics Club

Co-Founder, Co-President

Isla Vista, CA

October 2024 - Present

Led over 200 members in data analysis/visualization, machine learning, and deep learning workshops using Python libraries. Spearheading and managing technical consulting projects with UCSB Division 1 Baseball/Basketball teams. Co-organized two-day hackathon with 200+ participants, accruing over \$30000 in event funding. Coordinated a guest speaker panel featuring leading sports analytics professionals.

PROJECTS

GM-Agents (May - June 2025) Python, CrewAI, Playwright, LangGraph, OpenAI API

- Developed a multi-agent, heuristic-driven system that automates and optimizes strategy-based decision making and roster management in the Basketball GM browser game
- Augmented browsing agent with object-oriented scaffolding for heuristics and game state evaluation, integrating human-in-the-loop assessment workflows to validate and optimize automated decision-making processes.

Video Streaming Packet Analysis (Nov - Dec 2024) Python, Pandas, Numpy, Matplotlib, Scikit-Learn, NetUnicorn

- Built an end-to-end ML pipeline that predicts video-chunk download times ($R^2 = 0.90$)
- Automated packet capture and feature engineering, then validated the model on the Stanford Puffer dataset alongside freshly captured traces from YouTube, Twitch, and Vimeo.

Touch Grass (May 2023) Node.js, React Native, Google Firebase, Google Maps API, TypeScript

- Developed location-centric social media mobile application for college students utilizing React Native, TypeScript, Node.js, and Firebase.
- Won award for "Best Use of Google Cloud" at SB Hacks IX for Firebase, Google Maps API integration.

SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, Java, C#, SQL, PHP, Bash, HTML, CSS

Frameworks/Libraries: Django, React Native, Express, Bootstrap, Spring, jQuery, Numpy, Scipy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, Pytorch, Streamlit

Other: PostgreSQL, MySQL, Linux, Git, GitHub, AWS, MongoDB, Agile, Docker, Kubernetes, CI/CD

LICENSES AND CERTIFICATIONS

Machine Learning Specialization June 2024; DeepLearning.AI, Stanford University, Coursera

Proficiency in regression, classification, neural networks, unsupervised learning, reinforcement learning.